



The Sustainability committee and environmental disclosure: International evidence

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Abstract

This paper contributes to the growing debate about the role of board-level sustainability committees, focusing on whether they leverage sustainability expertise for impactful environmental initiatives or purely serve as symbolic actions aimed at greenwashing. Using a large set of firms from 35 countries over the 2010–2017 period, we find that the presence of a sustainability committee is positively associated with higher-quality GHG emissions disclosure. This finding is robust to endogeneity and sample selection bias concerns. The sustainability committee effect is more pronounced when external environmental institutions are too weak to properly monitor corporate environmental disclosure. Our findings suggest that sustainability committees are not a symbolic management tool, but play a crucial role in enhancing corporate environmental disclosure.

Introduction

The board of directors provides direction and oversight to implement strategies aimed at long-term value creation, recognizing that maximizing shareholder value is ultimately intertwined with protecting the interests of all stakeholders. The board's effectiveness in accomplishing this task depends on its governance structure. Eccles et al. (2014) document that firms voluntarily adopting sustainability policies are more likely to establish separate board committees to strengthen stakeholder engagement practices. According to the Sustainability Board Report (2021), more than

70 % of the world's largest 100 publicly listed companies have established a sustainability committee.² However, rather than having a substantive effect on a firm's sustainable efficiency, the sustainability committee may function merely to take symbolic actions to meet stakeholders' expectations and improve public perception as part of a "greenwashing" policy (Berrone and Gomez-Mejia, 2009; Peters et al., 2019). In this paper, we examine whether a board sustainability committee makes substantive use of sustainability-related board expertise or is only an example of "tokenism" to mask a lack of substantive sustainability policies and activities.

We tackle this research question from the perspective of firms' scope and quality of greenhouse gas (GHG) emissions reporting. In particular, we test whether the presence of a board sustainability committee leads to higher-quality GHG emissions disclosure.³ Taking this new focus is important for two main reasons. First, while some prior studies already document a positive impact of sustainability committees on firms' overall CSR reporting (Amran et al., 2014; Hussain et al., 2018; Gallego-Alvarez and Pucheta-Martinez, 2020), other studies find no such effect (Michelon and Parbonetti, 2012; Elsayih et al., 2018) and conclude that sustainability committees are only symbolic initiatives. To the best of our knowledge, no large-scale cross-country empirical evidence exists in the literature regarding the impact of sustainability committees on the scope and quality of GHG emissions disclosure, highlighting the need for further research to fill this gap.

Second, stakeholders increasingly pressure firms to evaluate the impact of their operations on climate change, particularly with respect to GHG emissions. The International Energy Agency (IEA, 2021) predicts that the carbon budget allowed to limit global warming to 1.5 °C will be exhausted within eight years. Any attempt to reduce corporate GHG emissions and achieve the "net zero" goal must start with measuring and reporting emissions. Referring to the financial disclosure literature (Leuz and Wysocki, 2016), Christensen et al. (2021) recommend that investors be concerned with environmental reporting in general, and GHG emissions reporting in particular. Better disclosure may lead to higher liquidity, lower financing costs, and higher firm value ("financial materiality"). Transparency about emissions can also induce changes in corporate behavior ("impact materiality") and lead to reductions in reporting firms' GHG emissions (Tomar, 2023).

In spite of these arguments, Bolton and Kacperczyk (2021) document that most firms still refuse to disclose their GHG emissions. Given the urgency of accelerating GHG emissions reductions, Bolton et al. (2021) recommend that governments mandate firms to report their direct emissions, i.e., scope 1 emissions that occur from sources controlled or owned by an organization.⁴ In contrast to mandatory corporate carbon disclosure and public enforcement, we propose the alternative argument that the structure of a firm's board of directors influences its willingness to disclose GHG emissions. Because boards of directors are responsible for advising on strategy development and monitoring its execution (Johnson et al., 1996), sustainability-conscious boards will also exert influence on firms' sustainability policies (Kassinis and Vafeas, 2002; Endrikat et al., 2021). One solution to establish institutionalized sustainability-related board expertise is to implement a board sustainability committee, a specialized board subgroup that deals with the creation,

implementation, and updating of environmental and social policies, the assessment of a firm's resource allocation decisions, and the coordination and monitoring of CSR-related issues (Gennari and Salvioni, 2019, p.763; Endrikat et al., 2021, p. 2108).

To legitimize activities that enhance environmental performance vis-à-vis shareholders and different stakeholders, firms should measure and disclose information about their GHG emissions levels. GHG emissions not only pose a risk to nature by accelerating climate change but also expose the emitting firm itself to risks (Flammer et al., 2021), including physical risk (e.g., droughts, extreme temperatures, and flooding), regulatory risk (e.g., emissions restrictions and pricing), and a variety of other risks (e.g., technological disruption, changing consumer preferences, and reputation threats). While firms face some of these risks irrespective of their own behavior (e.g., natural catastrophes), other risks are directly linked to their environmental actions (e.g., stakeholder dissatisfaction and reputation threats). High GHG emissions levels are also associated with higher likelihoods of compliance costs, fines, liabilities, litigations, and costs to adapt to future legislation as well as mandatory environmental standards (Matsumura et al., 2014; Griffin et al., 2017; Bolton and Kacperczyk, 2021). As a result of their high environmental risk exposure, the capital market trades high-polluting firms at a discount (Matsumura et al., 2014; Clarkson et al., 2015; Griffin et al., 2017; El Ghoul et al., 2018).

While many firms cannot attenuate these risks by reducing GHG emissions because of the nature of their business operations and often missing capabilities, a firm could attempt to reduce the associated risks (and the resulting valuation discount) by voluntarily disclosing its emissions. Without a mandatory disclosure rule, a higher scope and quality of voluntary GHG emissions disclosure helps stakeholders evaluate the consequences a firm potentially faces over its emissions. Our main premise is that GHG emissions disclosure enables more informed interactions between a firm and its stakeholders. This may, in turn, induce managers to incorporate emissions data into the firm's risk management and strategic plans to better handle and eventually reduce climate-related risks in the longer run, thereby preventing exaggerated stakeholder reactions and adverse effects on firm value. Given their pivotal role in designing environmental initiatives and introducing best sustainability management practices, we hypothesize that sustainability committees encourage and advise managers to prepare high-quality GHG emission reports. Lack of evidence to support this prediction could be interpreted as consistent with potential symbolic motivations.

To test our hypothesis on the relation between the presence of a sustainability committee and the firm's environmental disclosure quality, we use voluntary GHG emissions disclosure data from the Carbon Disclosure Project (CDP). We construct a large sample of firms from 35 countries over the 2010–2017 period and use aggregated measures capturing two aspects of environmental disclosure quality. The first measure relates to the level of detail at which firms report their emissions; the second denotes whether reported emissions have been externally verified. Our ordered logit regression results indicate that the presence of a sustainability committee has a positive and statistically significant impact on the quality of GHG emissions disclosure. Firms with a

sustainability committee disclose more detailed and reliable information about their carbon emissions. For example, firms with a sustainability committee are 26.4 percentage points more likely to choose the highest level of environmental disclosure than those without such a committee. These findings remain robust when controlling for the level of environmental performance, the presence of a chief sustainability officer, various board characteristics, and country-level variables such as stakeholder orientation and natural disasters. Overall, our results confirm the pivotal role of sustainability committees in disseminating detailed and reliable environmental information to corporate outsiders. In other words, sustainability committees have a real economic function and are not merely used as a symbolic management tool.

We use several approaches to address endogeneity concerns. First, we demonstrate that our findings are robust to including alternative fixed effects, such as firm, industry-year, or country-year fixed effects. Second, to ensure that differences in firm characteristics do not confound our estimates, we match each treated firm in our sample (with a sustainability committee) to one control firm (without a sustainability committee) from the same industry and country and with similar environmental performance, size, and growth profiles. Matched sample regressions confirm that a sustainability committee's presence improves the level and reliability of environmental disclosure. Third, we implement an instrumental variable approach and use major board reforms from Fauver et al. (2017) as an exogenous event that raises the likelihood of establishing a board-level sustainability committee but has no direct effect on environmental disclosure except through its impact on the sustainability committee and its actions. Consistent with a causal impact, we find that the instrumented sustainability committee has a positive and statistically significant effect on the level and reliability of environmental disclosure.

A further concern could be the non-random nature of our sample, which includes only firms that the CDP has invited on behalf of institutional investors to make their GHG emissions data public. Using the full sample instead, we additionally estimate an ordered probit sample selection model and find that sample selection bias does not drive our main evidence.

In our final set of tests, we examine whether the effectiveness (or lack thereof) of external environmental institutions affects the sustainability committee–carbon emissions reporting relationship. We find that the effect of a sustainability committee on the quality of GHG emissions disclosure is stronger when external environmental governance mechanisms (measured by, e.g., confidence in environmental organizations) are weaker or less effective.

Our study contributes to the literature on non-financial reporting, of which climate risk is the most urgent component (see Christensen et al. (2021) for a review). Flammer et al. (2021), Döring et al. (2023), and Ilhan et al. (2023) analyze the role of institutional investors in voluntary climate risk disclosure. We focus on the impact of sustainability committees on GHG emissions reporting and explore whether their presence enhances the scope and quality of disclosure or potentially masks greenwashing. Empirical studies investigating how sustainability committees affect GHG emissions disclosure are scant and inconclusive. Jaggi et al. (2018) and Helfaya and Moussa (2017) document a

positive relation between sustainability committees and environmental reporting using Italian and U.K. data, respectively, while Rankin et al. (2011) find no relation. We advance these earlier studies by providing compelling new cross-country evidence that sustainability committees improve the scope and quality of GHG emissions disclosure, especially when external environmental governance mechanisms are weak.

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Section snippets

Board sustainability committees

Board committees have gained popularity with the spread of corporate governance principles and best practices. Three committees are common in many countries: audit, nomination, and compensation. Adams et al. (2021) report that the proportion of board activities performed at the committee level in S&P 1500 firms increased from 36 % before to 52 % after the implementation of Sarbanes–Oxley. Over the last decade, increasing demand for managing stakeholder oversight has led to establishing another ...

Testable hypotheses

As already noted, sustainability committees can have various duties. They provide advice through informal knowledge sharing or formal reports (Radu and Samili, 2022) and encourage the entire board to develop opportunities and execute actions that generate and protect shareholder value (De Kluyver, 2009). In addition, sustainability committees operate as a control mechanism to prevent downside risk from any irresponsible firm behavior (Eccles et al., 2014; Burke et al., 2019) and possess the ...

Sample construction

In a perfect world, investors can devise their portfolio's carbon footprint. However, Amel-Zadeh and Serafeim (2018) conclude that reported information related to sustainability in general, and GHG emissions data in particular, lacks reliability and thus does not allow informed investment decisions. To ensure that firms inform stakeholders about their emissions meaningfully and

consistently, several initiatives have adopted standards aimed at improving emissions transparency. One prominent ...

Main evidence

Panel A of Table 4 presents the estimation results of various specifications of Eq. (1). Columns (1)–(4) ((5)–(8)) report the results using disclosure scope (emissions verification) as the dependent variable. In column (1), we focus on the effect of the presence of a sustainability committee on the scope of GHG emissions disclosure while omitting all the control variables. The coefficient estimate on sustainability committee loads positively and significantly, suggesting that firms with a ...

Conclusion

Disclosure of important information is the underpinning of functioning capital markets. Given the urgency of the climate crisis, GHG emissions information continues to grow in importance. Bolton et al. (2021) note that a requirement to report corporate carbon emissions could be fundamental to achieving net zero emissions. In this paper, we examine whether board sustainability committees leverage expertise to enhance the quality of firms' GHG emissions disclosure or whether they serve symbolic ...

Declaration of competing interest

None. ...

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References (85)

R.B. Adams *et al.*

[Death by committee? An analysis of corporate board \(sub-\) committees](#)

J. Financ. Econ. (2021)

N. Boubakri *et al.*

[Cross-listing and corporate social responsibility](#)

J. Corpor.Financ. (2016)

C.H. Cho *et al.*

[The role of environmental disclosures as tools of legitimacy: a research note. Accounting Organ. Soc. \(2007\)](#)

A. Dechezleprêtre *et al.*

Searching for carbon leaks in multinational companies

J. Environ. Econ. Manage (2022)

R. Duchin *et al.*

When are outside directors effective?

J. Financ. Econ. (2010)

S. El Ghoul *et al.*

Does corporate social responsibility affect the cost of capital?

J. Bank. Financ. (2011)

J. Elsayih *et al.*

Corporate governance and carbon transparency: australian evidence

Account. Res. J. (2018)

R. Fahr *et al.*

Who follows the crowd—Groups or individuals?

J. Econ. Behav. Organ. (2011)

O. Faleye *et al.*

The costs of intense board monitoring

J. Financ. Econ. (2011)

L. Fauver *et al.*

Board reforms and firm value: worldwide evidence

J. Financ. Econ. (2017)



View more references

Cited by (52)

How sustainability committees moderate the link between ESG performance and environmental innovation in European firms?

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2025, International Review of Financial Analysis

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The effectiveness of board gender diversity and sustainability committees in mitigating ESG controversies

2025, Meditari Accountancy Research

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Sustainability-related corporate governance and greenwashing practices: preliminary evidence from southeast Asian companies

2025, Meditari Accountancy Research

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2024, Journal of Corporate Finance

[Show abstract](#) 

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2026, Environment Development and Sustainability



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